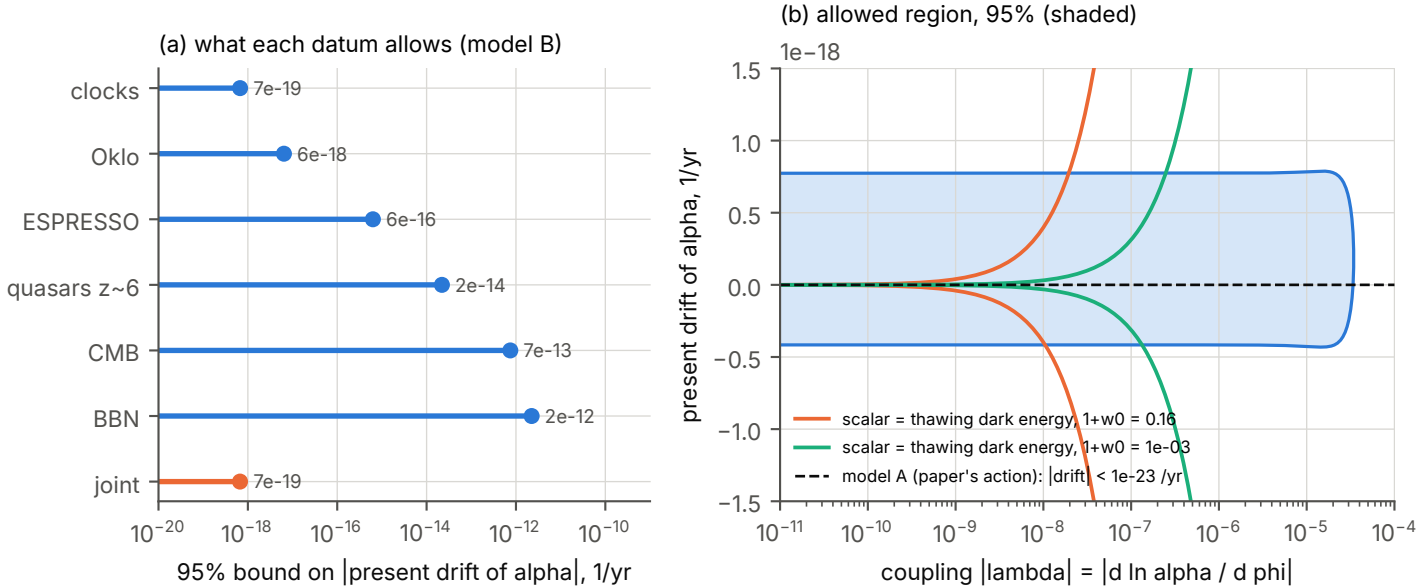
(a) what each datum allows (model B)
clocks
Oklo
ESPRESSO
quasars z~6
CMB
BBN
joint
7e-19
6e-18
6e-16
2e-14
7e-13
2e-12
7e-19
10^-20
10^-18
10^-16
10^-14
10^-12
10^-10
95% bound on |present drift of alpha|, 1/yr
(b) allowed region, 95% (shaded)
1e−18
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
present drift of alpha, 1/yr
scalar = thawing dark energy, 1+w0 = 0.16
scalar = thawing dark energy, 1+w0 = 1e-03
model A (paper's action): |drift| < 1e-23 /yr
10^-11
10^-10
10^-9
10^-8
10^-7
10^-6
10^-5
10^-4
coupling |lambda| = |d ln alpha / d phi|